

No. of Functional Features Included in the Solution

Date	26 September 2026
Team ID	SWTID-2026-2311
Project Name	AI StudyBuddy API
Maximum Marks	4 Marks

Functional Features Delivered

S.No	Feature	Description	Endpoint(s)
1	Secure User Authentication	Register, login and logout with bcryptjs-hashed passwords and access/refresh JWT issuance	POST /api/auth/register , /login , /logout
2	Role-Based Access Control	Student and Administrator roles enforced via protect and adminOnly middleware	Applies across protected/admin routes
3	Study Material Upload & Management	Upload, store and retrieve study material (validated file types and size) as input for AI processing	POST /api/materials/upload
4	AI Summary Generation	Google Gemini generates a concise summary from the uploaded material	POST /api/materials/:id/summarize
5	AI Flashcard Generation	Google Gemini generates question/answer flashcards from the uploaded material	POST /api/materials/:id/flashcards
6	AI Quiz Generation	Google Gemini generates multiple-choice quiz questions from the uploaded material	POST /api/materials/:id/quiz
7	AI Study Plan Generation	Google Gemini generates a personalized study plan from the material and the user's exam timeline	POST /api/materials/:id/study-plan
8	Administrative Management	Administrators manage registered users and monitor application/AI usage	/api/admin/*
9	Centralized Error Handling	Consistent structured JSON error responses for auth failures, validation errors and unexpected errors	Applies across all routes

9 functional features delivered within the 15–29 September 2026 cycle, matching the Key Features enumerated in the team's project write-up.